

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 3.2: Current Electricity — Teacher Reference (Pre-filled)

SUMMARY TABLE: PHYSICS GRADE 10	
Sub-Strand	Sub-Strand 3.2: Current Electricity
Driving Question	DRIVING QUESTION: Why do electrical devices in our homes sometimes receive less energy than expected, and how can we design circuits to deliver the right amount of electrical energy safely and efficiently?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Flickering Lights of Huruma Estate: Launching the Phenomenon	During a power restoration in Huruma Estate, Nairobi, lights glowed dimly, a kettle took longer to boil, and a bulb on a long thin extension cord was always dimmer than one plugged directly into the wall socket.	Electric current is the flow of electric charge through a conductor. Current I equals charge Q divided by time t , and is measured in amperes using an ammeter connected in series.	We cannot yet fully explain why the lights were dim or why the extension cord bulb was dimmer. We have identified that the AMOUNT of current flowing must be different in these situations, and we have posted our questions on the Driving Question Board to guide our investigation over the next 11 lessons.
Lesson 2 EMF and Potential Difference: Why Does the Socket Give Less Than the Source?	When a dry cell is connected to a resistor and the switch is closed, the voltmeter reading across the cell terminals drops below the open-circuit reading. The drop is larger when a smaller resistor (higher current) is used.	The energy given to each coulomb of charge by a source is called electromotive force (EMF, symbol E , unit volt). The energy available per coulomb in the external circuit is called terminal voltage or potential difference (symbol V). Terminal voltage is always less than or equal to EMF because some energy is used inside the source. Both quantities are measured in volts (joules per coulomb) using a voltmeter connected in parallel.	During the Huruma Estate power restoration, the electricity supply had a certain EMF but the terminal voltage reaching the wall sockets was lower than expected, especially when many appliances were drawing current simultaneously. This is why lights received less energy per coulomb than normal and glowed dimly. The voltage available at the socket is not the same as the rated supply EMF — it is reduced by losses inside the source, which become larger as current demand increases.
Lesson 3 Ohm's Law Investigation:	Students observed that as the rheostat was adjusted to increase the voltage across the nichrome wire, the current through the wire	Students learned that the gradient of a $V-I$ graph gives the resistance (R) of the conductor, measured in ohms (Ω). This	Students can now partially explain the Huruma Estate phenomenon: the long, thin extension cord has a higher resistance than

Resistance as the Controller of Current	also increased. When they plotted these values on a V-I graph, the points fell on a straight line passing through (or near) the origin — showing that voltage and current are directly proportional for the nichrome wire at constant temperature.	relationship is summarised by Ohm's Law: $V = IR$. Resistance is a property of the conductor that controls how much current flows for a given voltage — a higher resistance means less current for the same voltage.	a short, thick wire. Because $V = IR$, for the same mains voltage, a higher resistance in the cord means less current reaches the bulb — so it glows dimmer. The kettle element's resistance similarly controls how much current heats the water. Resistance is the missing variable that explains why 'electricity' can vary in its effect even from the same source.
Lesson 4 Resistance and Resistivity: Why Do Thicker Cables Carry More Current?	Students observed that when wire length increases (with area and material fixed), resistance increases proportionally. When cross-sectional area increases (with length and material fixed), resistance decreases. Different materials (copper vs. nichrome) gave different resistance values for identical wire dimensions. A thin, long wire had significantly higher resistance than a short, thick wire of the same material.	Students learned that resistance depends on three factors: length ($R \propto L$), cross-sectional area ($R \propto 1/A$), and material (characterised by resistivity ρ). These are combined in the formula $\rho = RA/L$ (units: $\Omega \cdot m$). Copper, used in Kenya Power-approved house wiring, has very low resistivity, making it an efficient conductor. Thin, cheap extension cords have much higher resistance due to their small cross-sectional area, which reduces current to appliances and causes energy to be wasted as heat in the cord.	Students explained that the neighbour's long, thin extension cord in Huruma Estate has high resistance because of its small cross-sectional area and long length (both increase R according to $\rho = RA/L$). This high cord resistance means the total circuit resistance is large, reducing the current. Less current means less power delivered to the bulb ($P = I^2R$), making it glow dimly even when the mains voltage is fully restored. This is why Kenya Power standards require thick-gauge wiring in homes — to minimise wire resistance and deliver maximum energy to appliances safely.
Lesson 5 Series Circuits: Same Current, Sharing Voltage — Why More Bulbs Mean Dimmer Light	When we added more bulbs to a single loop (series circuit) and measured with the ammeter, the current reading was the same at every point in the circuit — before the first bulb, between bulbs, and after the last bulb. The bulbs became progressively dimmer as we added more bulbs to the series loop. When we summed the voltage readings across each individual bulb, the total equalled the source voltage.	In a series circuit: (1) current is the same everywhere in the loop; (2) the source voltage is shared (divided) among the components so that $V_{\text{source}} = V_1 + V_2 + V_3$; (3) resistances add up so that $R_{\text{total}} = R_1 + R_2 + R_3$; (4) adding more components increases total resistance, which reduces current and causes each component to receive less voltage and less power — making bulbs dimmer and heaters slower.	The neighbour in Huruma Estate whose bulb is connected via a long, thin extension cord has — without knowing it — created a series circuit. The resistance of the long thin cord adds to the resistance of the bulb in series, increasing the total resistance, reducing the current flowing through the bulb, and reducing the voltage across the bulb. Less current through a higher-resistance bulb means less power delivered to the filament, so the bulb glows dimmer. This is the same reason Christmas-light bulbs dim when more bulbs are added to the string — every new bulb is another series resistor sharing the fixed source voltage and increasing total resistance.
Lesson 6 Parallel Circuits: Why Every Appliance in a Kenyan Home Gets Full Voltage	Students built two-branch and three-branch parallel circuits using battery packs, bulbs, ammeters, and voltmeters. They measured that voltage across each parallel branch equalled the battery voltage (approximately	Students derived the three rules of parallel circuits: (1) voltage is the same across every branch and equals the supply voltage; (2) total current is the sum of all branch currents; (3) equivalent resistance is	Students can now explain one key part of the Huruma Estate phenomenon: inside the house, all appliances (lights, kettle) are wired in parallel and each receives the full supply voltage independently — which is

	6 V in each branch), that unscrewing one bulb left the other unaffected, and that total current in the main line equalled the sum of branch currents. They also calculated that equivalent resistance decreased as more branches were added.	calculated using $1/R_{eq} = 1/R_1 + 1/R_2 + \dots$ and is always less than the smallest individual branch resistance. Students connected these rules to Kenyan domestic wiring: all appliances are connected in parallel at 240 V so each operates independently at full voltage, and adding more appliances increases total current drawn from the Kenya Power supply.	why, once voltage is fully restored, the lights and kettle all operate normally and do not affect each other. Students also explained why the neighbour's bulb on a long thin extension cord is always dimmer: the extension cord adds a series resistance within that one parallel branch, reducing current and power in that branch alone, while all other parallel branches in the house remain unaffected.
Lesson 7 Series-Parallel Combination Circuits: Solving Real Kenyan Household Wiring Puzzles	In the Huruma Extension Cord Circuit, the voltage across the parallel appliance branch was approximately 208.7 V, not the full 240 V, because approximately 31.3 V was dropped across the 6-ohm extension cord. The bulb in Socket A and the charger in Socket B shared the reduced voltage, not the full supply voltage.	When a circuit contains both series and parallel sections, the series elements reduce the voltage available to the parallel load. The correct analysis method is step-by-step equivalent resistance reduction: first combine elements sharing the same two nodes (parallel rule), then combine sequential groups (series rule), then use Ohm's Law at each stage to find voltages and currents. Total current from the supply, voltage drop across the series element, and voltage across the parallel branch can all be calculated systematically.	The neighbour in Huruma Estate whose bulb is connected via a long thin extension cord always has a dimmer bulb because the cord acts as a series resistor in a combination circuit. It drops voltage before the current reaches the bulb, so the bulb receives less than the full 240 V supply voltage. This is not a fault in the supply — it is the predictable behaviour of a combination circuit where any series element steals voltage from the rest of the circuit.
Lesson 8 Internal Resistance and EMF: Why Does the Battery Weaken Under Load?	We observed that as we increased current drawn from the battery by decreasing external resistance, the voltmeter reading across the battery terminals fell steadily. The V versus I graph was a straight downward-sloping line, with the y-intercept equal to the open-circuit voltage (EMF) and a negative gradient.	We learned that every real battery has an internal resistance r that causes a voltage drop of Ir inside the battery whenever current flows. The terminal voltage V equals the EMF E minus the lost volts Ir : $V = E - Ir$. The EMF is the maximum voltage the source can provide (at zero current); the terminal voltage is always less when current is flowing.	We can now explain why the Huruma Estate lights flickered dimly when power was restored: at that moment, many appliances switched on simultaneously, drawing a large surge current I . This produced a large Ir drop inside the supply, reducing the terminal voltage at the sockets below the normal 240 V, so bulbs received less voltage and glowed dimly. We also explained the weakening torch: as the battery ages, its internal resistance r increases, so the Ir drop grows and terminal voltage falls even though EMF may remain roughly constant.
Lesson 9 Electrical Power: Deriving and Applying $P = IV = I^2R = V^2/R$	Students observed that: (1) a thin nichrome wire connected in a circuit became warm while a thick copper wire did not, even with the same supply voltage — and voltmeter/ammeter readings confirmed higher resistance and lower current in the nichrome circuit; (2) real Kenyan household device labels (electric jiko, immersion	Students learned that: (1) electrical power $P = IV = I^2R = V^2/R$ is the rate of energy transfer in a circuit, measured in watts; (2) power dissipated in a resistive conductor (such as an extension cord) increases with the SQUARE of current, meaning even a small increase in current causes a large increase in heating; (3) a long, thin	Students can now explain: (1) why the neighbour's bulb on a long, thin extension cord is always dimmer — the cord dissipates power as heat ($P = I^2R$), reducing the power available to the bulb; (2) why the electric kettle in Huruma Estate takes longer to boil during low-voltage periods — lower voltage means lower power delivered

	heater, phone charger) carry wattage ratings that can be used to calculate the current they draw; (3) their own Lesson 6 data tables, when extended with a 'Power (W)' column, showed that components with higher resistance dissipated more power for the same current.	extension cord with high resistance wastes a significant fraction of the total power as heat, leaving less power for the connected device — this is the quantitative explanation for the dimmer bulb in the Huruma Estate phenomenon; (4) electrical energy consumed is calculated as $E = Pt$, and is measured commercially in kilowatt-hours (kWh) by Kenya Power.	($P = V^2/R$), so energy is transferred to the water more slowly; (3) why undersized extension cords and overloaded circuits are genuine fire hazards — the power dissipated as heat in the wire can ignite insulation; (4) how to read a device label and calculate its operating current and resistance, and how to estimate a household's monthly electricity cost using the Kenya Power tariff.
Lesson 10 Electrical Energy and Kenya Power Billing	Students examined a sample Kenya Power electricity bill for a fictional Huruma Estate household and an appliance data table showing power ratings and daily usage times for six common Kenyan household devices including an electric jiko, kettle, immersion heater, and phone charger.	Electrical energy $E = Pt$, where energy in kilowatt-hours equals power in kilowatts multiplied by time in hours; one kilowatt-hour (1 kWh) is one unit on a Kenya Power bill charged at approximately Ksh 12.00 on the domestic lifeline tariff; the total cost of running an appliance depends on BOTH its power rating AND the duration of use; the Huruma Estate kettle consumed more total energy during the power restoration event because lower terminal voltage reduced the power delivered, requiring a longer running time to boil the same water.	The Huruma Estate kettle took longer to boil because the lower terminal voltage during the power surge reduced the effective power delivered to the heating element — less power per second meant less energy transferred to the water per minute, so more minutes were needed to transfer enough thermal energy to reach boiling point, and this longer running time also meant greater total electrical energy consumed and greater cost on the Kenya Power bill.
Lesson 11 Heating Effect of Electric Current & Electrical Safety	Students observed that when current flowed through nichrome wire coiled around a water-filled test tube, the water temperature increased — and the increase was greater at higher voltage (higher current). They observed that nichrome wire became warm while copper connecting wires carrying the same current stayed cool. They examined photographs of a charred extension cord from Huruma Estate and appliance rating plates for Kenyan household devices (Ramtons kettle, Sayona cooker, Nunix immersion heater).	Students learned that electrical energy is converted to heat energy in any resistor carrying current, quantified by $Q = I^2Rt$ (Joule's Law). High-resistance elements (nichrome, toaster coil, kettle element) convert more electrical energy to heat than low-resistance conductors (copper wires), which is why heating appliances use high-resistance alloys deliberately. During the Huruma brownout, lower voltage reduced current ($I = V/R$), which dramatically reduced heat output ($Q \propto I^2$), explaining why the kettle was slow. The long, thin extension cord of the neighbour had high resistance, so it dissipated significant heat — making it both a dim-bulb cause AND a fire hazard. Fuses (melt and break the circuit), circuit breakers (trip electromagnetically), and earthing (divert fault current safely to ground) are safety devices that prevent dangerous overcurrents from causing fires or electrocution.	The Huruma Estate phenomenon is now more fully explained: when power was first restored after the Kenya Power blackout, the supply voltage was below the standard 240 V (brownout condition). Lower voltage → lower current through each appliance → less power delivered ($P = V^2/R$) → less heat generated ($Q = I^2Rt$) → dimmer lights and a slower kettle. The neighbour's long, thin extension cord has high resistance, so even at full voltage it drops a significant portion of the supply voltage across itself ($V = IR$), reducing voltage across the bulb and wasting energy as heat in the cord. This same cord, if overloaded, could reach ignition temperatures — the fuse in the circuit is designed to melt and break the circuit before this happens, protecting the household from fire.

Lesson 12 Model Building, DQB Resolution & Final Explanation: Designing Safe and Efficient Circuits for Kenyan Homes	Students reviewed their Lesson 1 circuit models and annotated them with red-pen corrections, participated in an Evidence Parade revisiting eight key data sets from Lessons 1–11 (including the Ohm's Law V-I graph, the internal resistance terminal voltage graph, the series vs. parallel ammeter readings, and the Kenya Power energy cost data), and completed a gallery walk of peer-designed household wiring plans for the Wanjiku Family Home in Githurai, Nairobi.	Students synthesised all sub-strand concepts into a unified Final Explanation: that the Huruma Estate dim lights and slow kettle at power restoration were caused by high effective internal resistance in the supply network reducing terminal voltage and therefore power delivered to appliances ($P = V^2/R$); that the permanently dim bulb on the long thin extension cord resulted from resistive voltage drop across the cord in series with the bulb, leaving less voltage — and therefore less power — across the bulb; and that domestic wiring must be in parallel so that every appliance receives the full 240 V supply voltage, operates independently, and benefits from the low effective resistance of parallel branches. Students also learned to calculate total household current draw, assess fuse safety, and estimate monthly energy costs using Kenya Power tariff rates.	Students explained the complete Huruma Estate anchoring phenomenon using the Claim-Evidence-Reasoning-Design (CERD) framework, referencing Ohm's Law ($V = IR$), terminal voltage ($V_t = EMF - Ir$), power equations ($P = IV = V^2/R = I^2R$), parallel circuit rules (same voltage across branches, total current as sum of branch currents), and safety principles (correct fuse rating, earthing of metal-cased appliances). They closed every Driving Question Board card with evidence-referenced answers and justified a complete parallel wiring design for a model Kenyan household — demonstrating mastery of the sub-strand's core competency: designing safe, efficient electrical circuits for everyday Kenyan life.
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ATTRIBUTION AND LICENSING

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